Rates of Change and Rectilinear Motion

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Level 2 · Medium Find the maximum rate of change of $f(x, y) = 3x^2y - y^3$ at the point $(1, 2)$, and determine the unit vector in which it occurs.

Compute gradient: $ablaf = \langle 6xy, 3x^2 - 3y^2 \rangle$. At $(1, 2)$: $ablaf(1, 2) = \langle 6(1)(2), 3(1)^2 - 3(2)^2 \rangle = \langle 12, 3 - 12 \rangle = \langle 12, -9 \rangle$. Maximum rate of change is the gradient magnitude: $|ablaf(1, 2)| = \sqrt{12^2 + (-9)^2} = \sqrt{144 + 81} = \sqrt{225} = 15$ The direction of maximum increase is the normalized gradient: $\mathbf{u} = \frac{ablaf}{|ablaf|} = \frac{1}{15}\langle 12, -9 \rangle = \left\langle \frac{4}{5}, -\frac{3}{5} \right\rangle$

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A continuous random variable X has probability density function $f(x) = kx(2 - x)$ for $0 \le x \le 2$ and $f(x) = 0$ elsewhere. Determine: (a) the constant k; (b) the median of X; (c) the variance $\mathrm{Var}(X)$.

1. Normalization condition: $\int_0^2 k(2x - x^2)\,dx = 1 \Rightarrow k\left[x^2 - \frac{x^3}{3}\right]_0^2 = k\left(4 - \frac{8}{3}\right) = k\left(\frac{4}{3}\right) = 1 \Rightarrow k = \frac{3}{4}$. 2. Median m: By visual symmetry of the parabola $f(x) = \frac{3}{4}x(2 - x)$ about $x = 1$, $m = 1$. We verify: $F(1) = \int_0^1 \frac{3}{4}(2x - x^2)\,dx = \frac{3}{4}\left[x^2 - \frac{x^3}{3}\right]_0^1 = \frac{3}{4}\left(\frac{2}{3}\right) = \frac{1}{2}$. ✓ 3. Variance: By symmetry, $E[X] = 1$..

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Express $z = -1 + i\sqrt{3}$ in polar form $r(\cos\theta + i\sin\theta)$ and exponential form $re^{i\theta}$, with principal argument $-\pi < \theta \le \pi$. Hence find z^6 and determine all three solutions to $w^3 = z$.

1. Modulus: $r = |z| = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{1 + 3} = 2$. Argument: since z lies in Quadrant II, $\theta = \pi - \arctan(\sqrt{3}) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$. Polar form: $z = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$; Exponential form: $z = 2e^{i2\pi/3}$. 2. Compute z^6 via De Moivre's Theorem: $z^6 = 2^6 e^{i \cdot 6(2\pi/3)} = 64e^{i4\pi} = 64(\cos 4\pi + i\sin 4\pi) = 64(1 + 0i) = 64$. 3. Solve $w^3 = z = 2e^{i2\pi/3}$.

4 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Use repeated integration by parts to evaluate the exact definite integral $\int_0^{\pi/2} x^2\cos x\,dx$.

1. First IBP: Let $u = x^2 \Rightarrow du = 2x\,dx$. Let $dv = \cos x\,dx \Rightarrow v = \sin x$. $\int_0^{\pi/2} x^2\cos x\,dx = [x^2\sin x]_0^{\pi/2} - 2\int_0^{\pi/2} x\sin x\,dx = \left(\left(\frac{\pi}{2}\right)^2(1) - 0\right) - 2\int_0^{\pi/2} x\sin x\,dx = \frac{\pi^2}{4} - 2J$. 2. Second IBP on $J = \int_0^{\pi/2} x\sin x\,dx$: Let $u = x \Rightarrow du = dx$. Let $dv = \sin x\,dx \Rightarrow v = -\cos x$..

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5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Give the first three nonzero terms of the Maclaurin series for $\sin x$.

Theorem (Maclaurin Series for Sine): The sine function is odd, so its Maclaurin series contains exclusively odd powers of x with alternating signs. Step 1 · Compute successive derivatives at $x = 0$: $f(x) = \sin x \Rightarrow f(0) = 0$ $f'(x) = \cos x \Rightarrow f'(0) = 1$ $f''(x) = -\sin x \Rightarrow f''(0) = 0$ $f'''(x) = -\cos x \Rightarrow f'''(0) = -1$ $f^{(4)}(x) = \sin x \Rightarrow f^{(4)}(0) = 0$ $f^{(5)}(x) = \cos x \Rightarrow f^{(5)}(0) = 1$ Step 2 · Form the first three nonzero terms.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Give the first three terms of the Maclaurin series for $\cos x$.

Theorem (Maclaurin Series for Cosine): The cosine function is even, so its Maclaurin series consists exclusively of even powers of x with alternating signs. Step 1 · General Maclaurin formula for $\cos x$: $\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$. Step 2 · Write out the terms for $n = 0, 1, 2$: $n = 0 \Rightarrow (-1)^0 \frac{x^0}{0!} = 1$ $n = 1 \Rightarrow (-1)^1 \frac{x^2}{2!} = -\frac{x^2}{2}$ $n = 2 \Rightarrow (-1)^2 \frac{x^4}{4!} = +\frac{x^4}{24}$ Conclusion: The first three terms are.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

(Paper 3 Extended Investigation) Consider the recursive Babylonian sequence defined by $x_0 = 3$ and $x_{n+1} = \frac{1}{2}\left(x_n + \frac{5}{x_n}\right)$ for $n \geq 0$. (a) Use the AM-GM inequality to prove that $x_n \geq \sqrt{5}$ for all $n \geq 1$. (b) Show that the sequence is strictly decreasing for $n \geq 1$, and deduce that $\lim_{n\to\infty} x_n = \sqrt{5}$. (c) Prove the quadratic error relationship $|x_{n+1} - \sqrt{5}| = \frac{(x_n - \sqrt{5})^2}{2x_n}$.

(a) Lower bound via AM-GM: For any positive real number $x_n > 0$, the Arithmetic Mean-Geometric Mean inequality states that $\frac{a+b}{2} \geq \sqrt{ab}$. With $a = x_n$ and $b = \frac{5}{x_n}$: $x_{n+1} = \frac{x_n + 5/x_n}{2} \geq \sqrt{x_n \cdot \frac{5}{x_n}} = \sqrt{5}$. Thus $x_n \geq \sqrt{5}$ for all $n \geq 1$. (b) Monotonicity & Convergence: Compute the difference: $x_{n+1} - x_n = \frac{1}{2}x_n + \frac{5}{2x_n} - x_n = \frac{5 - x_n^2}{2x_n}$. Since $x_n \geq \sqrt{5}$ for $n \geq 1$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Two straight paths in 3D space are modeled by lines $L_1 : \mathbf{r}_1 = (1; 2; 0) + \lambda(1; -1; 2)$ and $L_2 : \mathbf{r}_2 = (0; 1; -1) + \mu(2; 1; 1)$. Show that L_1 and L_2 are skew, and find the shortest perpendicular distance between them.

1. Non-parallel test: Direction vectors $\mathbf{d}_1 = \langle 1, -1, 2 \rangle$ and $\mathbf{d}_2 = \langle 2, 1, 1 \rangle$ are not scalar multiples ($1/2eq -1/1$). 2. Normal direction perpendicular to both lines:

$\mathbf{n} = \mathbf{d}_1 \times \mathbf{d}_2 = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 1, -1, 2; 2, 1, 1| = \mathbf{i}(-1-2) - \mathbf{j}(1-4) + \mathbf{k}(1-(-2)) = -3\mathbf{i} + 3\mathbf{j} + 3\mathbf{k} \propto (-1; 1; 1).$ Magnitude.

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9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

For what x does $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ converge?

Theorem (Ratio Test for Absolute Convergence): For an infinite series $\sum a_n$, evaluate $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$. The series converges absolutely when $L < 1$. Step 1 · Set up the ratio for $a_n = \frac{x^n}{n!}$:

$L = \lim_{n\to\infty} \left|\frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n}\right| = \lim_{n\to\infty} \left|\frac{x \cdot x^n}{(n+1)\cdot n!} \cdot \frac{n!}{x^n}\right| = \lim_{n\to\infty} \frac{|x|}{n+1}$. Step 2 · Evaluate the limit f.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

The polar curve $r = 2\cos\theta$ is a circle. Find its enclosed area.

Theorem (Area of Circle in Polar Coordinates): The curve $r = 2\cos\theta$ is a circle passing through the pole. Multiplying by r yields $r^2 = 2r\cos\theta \Rightarrow x^2 + y^2 = 2x \Rightarrow (x-1)^2 + y^2 = 1$, which is a circle of radius $R = 1$ centered at $(1, 0)$. Step 1 · Identify integration bounds: The circle is completely traced as θ goes from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ (or from 0 to π). Step 2 · Set up and evaluate the polar integral.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 3 · AdvancedFind the coordinates of all points on the curve $x^2 - xy + y^2 = 9$ where the tangent line is horizontal.

Step 1 · Implicit Differentiation: Differentiate with respect to x:

$2x - \left(1\cdot y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0 \Rightarrow 2x - y + (2y - x)\frac{dy}{dx} = 0$. Solve for $\frac{dy}{dx}$: $\frac{dy}{dx} = \frac{y - 2x}{2y - x}$. Step 2 · Condition for Horizontal Tangent: Set the numerator equal to zero while ensuring the denominator is non-zero:

$y - 2x = 0 \Rightarrow y = 2x$. Step 3 · Substitute into Original Curve.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Use De Moivre's theorem to prove that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$. Hence solve $4x^3 - 3x = 0$ on $x \in [-1, 1]$.

1. Expand $(\cos\theta + i\sin\theta)^3$ via Binomial Theorem: $(\cos\theta + i\sin\theta)^3 = \cos^3\theta + 3i\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - i\sin^3\theta$. 2. By De Moivre's Theorem, this equals $\cos 3\theta + i\sin 3\theta$. Equate real parts: $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$. Substitute $\sin^2\theta = 1 - \cos^2\theta$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

(Paper 3 Extended Investigation) Define the integral family $I_n = \int_0^1 x^n e^x\,dx$ for integers $n \geq 0$. (a) Evaluate I_0. (b) Use integration by parts to establish the reduction formula $I_n = e - nI_{n-1}$ for $n \geq 1$. (c) Hence determine the exact value of $\int_0^1 x^3 e^x\,dx$. (d) Prove that $0 < I_n < \frac{e}{n+1}$, and deduce $\lim_{n\to\infty} I_n$.

(a) Base integral: $I_0 = \int_0^1 e^x\,dx = [e^x]_0^1 = e^1 - e^0 = e - 1$. (b) Reduction formula via IBP: Let $u = x^n \Rightarrow du = nx^{n-1}\,dx$. Let $dv = e^x\,dx \Rightarrow v = e^x$. $I_n = [x^n e^x]_0^1 - n\int_0^1 x^{n-1}e^x\,dx = (1^n e^1 - 0) - nI_{n-1} = e - nI_{n-1}$. ■ (c) Compute I_3: - $I_1 = e - 1(I_0) = e - (e-1) = 1$. - $I_2 = e - 2(I_1) = e - 2(1) = e - 2$. - $I_3 = e - 3(I_2) = e - 3(e-2) = e - 3e + 6 = 6 - 2e$. (d) Squeeze Theorem & Limit: For.

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14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

For the rational function $f(x) = \frac{x^2 - x - 2}{x - 3}$: (a) find the equations of all asymptotes; (b) find the coordinates of any stationary points and determine their nature.

(a) Vertical Asymptote: The denominator is zero at $x = 3$. Numerator at $x = 3$ is $3^2 - 3 - 2 = 4eq0$. Thus $x = 3$ is a vertical asymptote. Oblique Asymptote: Perform polynomial long division: $\frac{x^2 - x - 2}{x - 3} = x + 2 + \frac{4}{x - 3}$. As $x \to \pm\infty$, $\frac{4}{x - 3} \to 0$. Therefore, $y = x + 2$ is the oblique asymptote. (b) Differentiate $f(x) = x + 2 + 4(x - 3)^{-1}$: $f'(x) = 1 - \frac{4}{(x - 3)^2}$. Set $f'(x) = 0$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule**

Use the substitution $x = 2\sin\theta$ to evaluate the exact definite integral $\int_0^1 \sqrt{4 - x^2}\, dx$.

1. Substitute $x = 2\sin\theta \Rightarrow dx = 2\cos\theta\, d\theta$. When $x = 0$, $\theta = 0$. When $x = 1$, $2\sin\theta = 1 \Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$. 2. Rewrite integrand: $\sqrt{4 - x^2} = \sqrt{4 - 4\sin^2\theta} = 2\sqrt{\cos^2\theta} = 2\cos\theta$ (since $\cos\theta > 0$ on $[0, \pi/6]$). 3. Substitute into the integral.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 5.01 Extrema and the Mean Value Theorem**

(IB AI HL) A market transition matrix $T = (0.7, 0.2; 0.3, 0.8)$ represents weekly customer movement between Brand A and Brand B. Find: (a) the probability that a customer initially with Brand A is with Brand B after 2 weeks; (b) the long-term steady-state market distribution.

(a) State distribution after 2 weeks is $\mathbf{p}_2 = T^2\mathbf{p}_0$. With $\mathbf{p}_0 = (1; 0)$: $T^2 = (0.7, 0.2; 0.3, 0.8)(0.7, 0.2; 0.3, 0.8) = (0.49 + 0.06, 0.14 + 0.16; 0.21 + 0.24, 0.06 + 0.64) = (0.55, 0.30; 0.45, 0.70)$. Thus $\mathbf{p}_2 = (0.55; 0.45)$, so the probability of being with Brand B after 2 weeks is 0.45 (45%). (b) Steady-state vector $\mathbf{s} = (s_1; s_2)$ satisfies $(T - I)\mathbf{s} = \mathbf{0}$.

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization**

Level 2 · MediumFind the point (x, y) on the parabola $y = x^2$ that is closest to the point $(0, 2)$.

Step 1 · Distance Objective: The distance d from (x, x^2) to $(0, 2)$ is: $d = \sqrt{(x - 0)^2 + (x^2 - 2)^2}$. Minimizing d is equivalent to minimizing the squared distance function $S(x) = d^2$: $S(x) = x^2 + (x^2 - 2)^2 = x^2 + x^4 - 4x^2 + 4 = x^4 - 3x^2 + 4$. Step 2 · Critical Numbers: $S'(x) = 4x^3 - 6x = 2x(2x^2 - 3) = 0$. Critical roots: $x = 0$ and $x = \pm\sqrt{\frac{3}{2}} = \pm\frac{\sqrt{6}}{2}$. Step 3 · Second Derivative Test: $S''(x) = 12x^2 - 6$. • At $x = 0$: $S''(0) = -60$ (certified loc.

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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Find all five solutions to $z^5 = -32$ in polar form $re^{i\theta}$ with $-\pi < \theta \le \pi$. Show that the sum of these five roots equals 0.

1. Write $-32 = 32e^{i\pi}$. By De Moivre's Theorem: $z_k = 32^{1/5}\exp\left(i\frac{\pi + 2\pi k}{5}\right) = 2e^{i(\pi + 2\pi k)/5}$ for $k = 0, 1, 2, 3, 4$. Evaluating the arguments in $(-\pi, \pi]$: - $k = 0$: $z_0 = 2e^{i\pi/5}$ - $k = 1$: $z_1 = 2e^{i3\pi/5}$ - $k = 2$: $z_2 = 2e^{i5\pi/5} = 2e^{i\pi} = -2$ - $k = 3$: $z_3 = 2e^{i7\pi/5} = 2e^{-i3\pi/5}$ - $k = 4$: $z_4 = 2e^{i9\pi/5} = 2e^{-i\pi/5}$ 2. Sum of the roots: Notice that.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Prove by mathematical induction that $5^n - 1$ is divisible by 4 for all positive integers $n \in \mathbb{Z}^+$.

Base Step: For $n = 1$: $5^1 - 1 = 4 = 4(1)$. Since 4 is divisible by 4, the statement $P(1)$ is true. Inductive Hypothesis: Assume $P(k)$ is true for some positive integer $k \ge 1$, meaning: $5^k - 1 = 4m$ for some integer $m \in \mathbb{Z} \Rightarrow 5^k = 4m + 1$. Inductive Step: Consider $n = k + 1$: $5^{k+1} - 1 = 5 \cdot 5^k - 1$. Substitute the inductive hypothesis $5^k = 4m + 1$: $5(4m + 1) - 1 = 20m + 5 - 1 = 20m + 4 = 4(5m + 1)$. Since $m \in \mathbb{Z}$, $5m + 1$ is an integer, so.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Level 3 · Advanced / AP FRQEvaluate the improper integral $\int_0^\infty xe^{-2x}\,dx$ or establish that it diverges.

Rewrite with limit notation: $\lim_{b\to\infty} \int_0^b xe^{-2x}\,dx$. Use integration by parts: $u = x \Rightarrow du = dx$, $dv = e^{-2x}\,dx \Rightarrow v = -\frac{1}{2}e^{-2x}$. $\int xe^{-2x}\,dx = -\frac{1}{2}xe^{-2x} - \int\left(-\frac{1}{2}e^{-2x}\right)dx = -\frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x}$. Evaluate from 0 to b: $\lim_{b\to\infty}\left[-\frac{1}{2}be^{-2b} - \frac{1}{4}e^{-2b} - \left(0 - \frac{1}{4}e^0\right)\right]$. By L'Hôpital's Rule.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A particle moves along the x-axis with position $x(t) = t^3 - 6t^2 + 9t$ for $t \ge 0$. At what times is the particle at rest?

a. $t = 1$ **and** $t = 3$ ✓

b. $t = 0$ and $t = 3$

c. $t = 2$

d. $t = 3$ only

At rest means velocity $v(t) = x'(t) = 0$: $3t^2 - 12t + 9 = 0 \Rightarrow 3(t - 1)(t - 3) = 0 \Rightarrow t = 1, 3$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the speed of the particle when $v(t) = -8$ and $a(t) = 3$?

a. 8 ✓

b. -8

c. 5

d. $\sqrt{73}$

Speed is defined as the absolute value of velocity: $|v(t)| = |-8| = 8$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

When is a particle's speed increasing?

a. **When velocity and acceleration have the same sign** ✓

b. When velocity is positive

c. When acceleration is positive

d. When acceleration is zero

Speed $|v(t)|$ increases when velocity and acceleration work in the same direction, meaning $v(t)$ and $a(t)$ have the same sign.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If a particle's position is $s(t) = t^2 - 4t$ on $[0, 3]$, what is its net displacement?

a. -3 ✓

b. 5

c. -4

d. 3

Displacement $= s(3) - s(0) = (9 - 12) - 0 = -3$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the above particle with $s(t) = t^2 - 4t$ on $[0, 3]$, what is the total distance traveled?

a. 5 ✓

b. 3

c. 4

d. 7

$v(t) = 2t - 4 = 0 \Rightarrow t = 2$. From $t = 0$ to 2: $|s(2) - s(0)| = |-4 - 0| = 4$. From $t = 2$ to 3: $|s(3) - s(2)| = |-3 - (-4)| = 1$. Total distance $= 4 + 1 = 5$.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which integral represents the total distance traveled by a particle from $t = a$ to $t = b$?

a. $\int_a^b |v(t)|dt$ ✓

b. $\int_a^b v(t)dt$

c. $|\int_a^b v(t)dt|$

d. $\int_a^b a(t)dt$

Total distance accumulates the speed over time: $\int_a^b |v(t)|dt$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The radius of a circle increases at a constant rate of 3 cm/s. How fast is the area increasing when the radius is 10 cm?

a. 60π **cm²/s** ✓

b. 30π cm²/s

c. 100π cm²/s

d. 20π cm²/s

$A = \pi r^2 \Rightarrow \frac{dA}{dt} = 2\pi r\frac{dr}{dt} = 2\pi(10)(3) = 60\pi$ cm²/s.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A 13-foot ladder leans against a vertical wall. If the bottom slides away from the wall at 2 ft/s, how fast is the top sliding down when the bottom is 5 feet from the wall?

a. $-\frac{5}{6}$ **ft/s** ✓

b. $-\frac{12}{5}$ ft/s

c. -2 ft/s

d. $-\frac{5}{12}$ ft/s

$x^2 + y^2 = 13^2 = 169$. When $x = 5$, $y = 12$. Differentiating:
$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow 5(2) + 12\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{10}{12} = -\frac{5}{6}$ ft/s.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Air is pumped into a spherical balloon at a rate of 100 cm³/s. How fast is the radius increasing when $r = 5$ cm?

a. $\frac{1}{\pi}$ **cm/s** ✓

b. $\frac{1}{4\pi}$ cm/s

c. $\frac{4}{\pi}$ cm/s

d. 25π cm/s

$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2\frac{dr}{dt} \Rightarrow 100 = 4\pi(25)\frac{dr}{dt} \Rightarrow 100 = 100\pi\frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{1}{\pi}$ cm/s.

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A conical tank with vertex down has height 10 m and base radius 4 m. Water flows in at 2 m^3/min. What is the relation between r and h?

a. $r = \frac{2}{5}h$ ✓

b. $r = \frac{5}{2}h$

c. $r = \frac{1}{2}h$

d. $r = 4h$

By similar triangles in the cross-section: $\frac{r}{h} = \frac{4}{10} = \frac{2}{5} \Rightarrow r = \frac{2}{5}h$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Using $V = \frac{1}{3}\pi r^2 h = \frac{4}{75}\pi h^3$, how fast is the water level rising when $h = 5$ m and $\frac{dV}{dt} = 2$ m^3/min?

a. $\frac{1}{2\pi}$ **m/min** ✓

b. $\frac{2}{\pi}$ m/min

c. $\frac{1}{4\pi}$ m/min

d. $\frac{4}{25\pi}$ m/min

$\frac{dV}{dt} = \frac{4}{25}\pi h^2 \frac{dh}{dt} \Rightarrow 2 = \frac{4}{25}\pi(25)\frac{dh}{dt} = 4\pi\frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{2}{4\pi} = \frac{1}{2\pi}$ m/min.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A street light is mounted at the top of a 15-foot pole. A 6-foot man walks away from the pole at 5 ft/s. How fast is the tip of his shadow moving?

a. $\frac{25}{3}$ **ft/s** ✓

b. 5 ft/s

c. $\frac{10}{3}$ ft/s

d. 10 ft/s

Let s be shadow tip distance: $\frac{s}{15} = \frac{s-x}{6} \Rightarrow 6s = 15s - 15x \Rightarrow 9s = 15x \Rightarrow s = \frac{5}{3}x$. Then $\frac{ds}{dt} = \frac{5}{3}\frac{dx}{dt} = \frac{5}{3}(5) = \frac{25}{3}$ ft/s.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Under the change of variables $x = 2u$, $y = -2u + 4v$, a region S in the uv-plane with area 2 maps to a region R. Find the area of R.

a. 2

b. 8

c. 12

d. 16 ✓

$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = |8|$, so $\text{Area}(R) = |J| \cdot \text{Area}(S) = 16$.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Related rates: a balloon radius expands at 2 cm/s. Rate of volume change when $r = 5$ cm is:

a. 40π cm³/s

b. 100π cm³/s

c. 200π **cm³/s** ✓

d. 300π cm³/s

$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(25)(2) = 200\pi$ cm³/s.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Related rates: ladder 10 m against wall. Base slides at 1 m/s when base is 6 m. Rate of top:

a. $-3/4$ **m/s** ✓

b. $3/4$ m/s

c. -1 m/s

d. $-4/3$ m/s

$x^2 + y^2 = 100 \Rightarrow 2xy' + 2x\dot{x} = 0 \Rightarrow y' = -6/(8)\cdot 1 = -3/4$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Change order: $\int_0^2 \int_0^y f\,dx\,dy$ becomes $\int_0^2 \int_?^? f\,dy\,dx$:

a. $\int_0^2 \int_x^1 f\,dy\,dx$

b. $\int_0^2 \int_0^x f\,dy\,dx$

c. $\int_0^1 \int_x^2 f\,dy\,dx$

d. $\int_0^2 \int_x^2 f\,dy\,dx$ ✓

Region $0 \le x \le y \le 2 \Leftrightarrow 0 \le x \le 2, x \le y \le 2$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The change-of-variables formula $\iint_D f\,dA = \iint_{D^*} f(g)\,|J|\,du\,dv$ requires $Je0$ (injective map). The factor $|J|$ is the:

a. Laplacian

b. Hessian

c. **Jacobian determinant** ✓

d. divergence

Jacobian: area scaling factor in change of variables.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Change of variables: $\iint_D f(x,y)\,dA = \iint_{D'} f(x(u,v), y(u,v))|J|\,du\,dv$ where $J =$

a. $x_u y_v + x_v y_u$

b. $x_u y_v - x_v y_u$ ✓

c. $x_u + y_v$

d. 1

Jacobian determinant: $J = x_u y_v - x_v y_u$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Consider the region R in the xy-plane bounded by the curves $y = x^2$ and $y = 4$. What is the area of this region?

a. $\frac{32}{3}$ ✓

b. $\frac{16}{3}$

c. 8

d. 16

The area of the region bounded by two curves $y = f(x)$ and $y = g(x)$ from $x = a$ to $x = b$, where $f(x) \geq g(x)$ on $[a, b]$, is given by the integral $\int_a^b (f(x) - g(x))dx$. First, we find the intersection points of $y = x^2$ and $y = 4$ by setting them equal: $x^2 = 4$, which gives $x = -2$ and $x = 2$. So, our limits of integration are $a = -2$ and $b = 2$. On the interval $[-2, 2]$, the line $y = 4$ is above the parabola $y = x^2$. Therefore, the area A is given by: $A = \int_{-2}^{2} (4 - x^2)dx$ Now, we eva.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Differential Equations and Slope Fields

The figure shows the region R bounded by the curves $y = x^2$ and $y = 4$. Which of the following double integrals represents the area of region R using the order of integration $dydx$?

a. $\int_{-2}^{2}\int_{x^2}^{4} dydx$ ✓

b. $\int_{-2}^{2}\int_{4}^{x^2} dydx$

c. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dydx$

d. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dxdy$

The region R is bounded above by $y = 4$ and below by $y = x^2$. To find the limits for x, we find the intersection points of the two curves: $x^2 = 4 \Rightarrow x = \pm 2$. Thus, the region is defined for x from -2 to 2. When integrating with respect to y first ($dydx$), the inner integral goes from the lower curve to the upper curve, so from $y = x^2$ to $y = 4$. The outer integral is over the range of x, which is -2 to 2. Therefore, the double integral representing the area is.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Area Between Curves

A particle's motion is described by the parametric equations $x(t) = e^{2t} - 2t$ and $y(t) = 4e^t$ for $t \geq 0$. Find the value of $\frac{dy}{dx}$ at the point where the tangent line to the curve is horizontal.

a. **0** ✓

b. 1

c. 2

d. 4

To find the slope of the tangent line, we compute $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$. \\ First, we find the derivatives with respect to t: \\ $\frac{dx}{dt} = \frac{d}{dt}(e^{2t} - 2t) = 2e^{2t} - 2$ \\ $\frac{dy}{dt} = \frac{d}{dt}(4e^t) = 4e^t$ \\ So, $\frac{dy}{dx} = \frac{4e^t}{2e^{2t} - 2}$. \\ A horizontal tangent line occurs when $\frac{dy}{dt} = 0$ and $\frac{dx}{dt} eq 0$. \\ Setting $\frac{dy}{dt} = 0$, we get $4e^t = 0$. This equation has no solution for real t since $e^t > 0$ for all real t. \\.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Consider the solid E bounded by the surfaces $z = x^2 + y^2$ and $z = 8 - x^2 - y^2$. Calculate the volume of this solid using a triple integral by first transforming to cylindrical coordinates. Express your answer in terms of π.

a. $\frac{16\pi}{3}$

b. $\frac{32\pi}{3}$ ✓

c. 16π

d. $\frac{64\pi}{3}$

The solid E is bounded below by the paraboloid $z = x^2 + y^2$ and above by the paraboloid $z = 8 - x^2 - y^2$. To find the intersection of these two surfaces, we set them equal: $x^2 + y^2 = 8 - x^2 - y^2$, which simplifies to $2(x^2 + y^2) = 8$, or $x^2 + y^2 = 4$. This is a circle of radius 2 in the xy-plane. In cylindrical coordinates, $x = r\cos(\theta)$, $y = r\sin(\theta)$, and $z = z$. The equations of the surfaces become $z = r^2$ and $z = 8 - r^2$. The intersection occurs at $r^2 = 8 - r^2$, so.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Limits Graphically and Numerically

The rate of change of a quantity $Q(t)$ is given by $\frac{dQ}{dt} = kte^{-0.5t^2}$, where k is a positive constant. If $Q(0) = 5$ and $Q(2) = 15$, find the value of k.

a. $\frac{5}{1 - e^{-2}}$

b. $\frac{5}{e^{-2} - 1}$

c. $\frac{10}{1 - e^{-2}}$ ✓

d. $\frac{10}{e^{-2} - 1}$

We are given the differential equation $\frac{dQ}{dt} = kte^{-0.5t^2}$. To find $Q(t)$, we need to integrate both sides with respect to t. The integral of the right side requires a substitution. Let $u = -0.5t^2$, so $du = -t\,dt$ or $-du = t\,dt$. Thus, $kte^{-0.5t^2}\,dt = -ke^u\,du$. Integrating $\int kte^{-0.5t^2}dt$ gives $\int -ke^u\,du = -ke^u + C = -ke^{-0.5t^2} + C$. So, $Q(t) = -ke^{-0.5t^2} + C$. We are given $Q(0) = 5$. Substituting $t = 0$ into the equation for $Q(t)$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

A population of bacteria follows a logistic growth model where $P(t)$ is the population at time t (in hours). The differential equation governing the population growth is $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{1000}\right)$. If the initial population is $P(0) = 200$, use Euler's method with a step size of $\Delta t = 1$ hour to approximate the population after 1 hour, $P(1)$.

a. 210

b. **208** ✓

c. 212

d. 209

Euler's method approximates the next value $P(t + \Delta t)$ using the formula $P(t + \Delta t) \approx P(t) + \frac{dP}{dt}|_t \cdot \Delta t$. We are given $P(0) = 200$ and $\Delta t = 1$. First, we find the rate of change at $t = 0$: $\frac{dP}{dt}|_{t=0} = 0.05 \cdot P(0)\left(1 - \frac{P(0)}{1000}\right)$ $\frac{dP}{dt}|_{t=0} = 0.05 \cdot 200\left(1 - \frac{200}{1000}\right)$ $\frac{dP}{dt}|_{t=0} = 10(1 - 0.2)$ $\frac{dP}{dt}|_{t=0} = 10 \cdot 0.8$ $\frac{dP}{dt}|_{t=0} = 8$ Now, we.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 The Definition of the Derivative

The rate of change of the volume of a spherical balloon being inflated is given by $\frac{dV}{dt} = 12\pi$ cm^3/s. If the initial volume of the balloon is 36π cm^3, find the volume of the balloon after t seconds. The volume V of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.

a. $V(t) = 12\pi t + 36\pi$ ✓

b. $V(t) = \frac{4}{3}\pi t^3 + 36\pi$

c. $V(t) = 4\pi t^3 + 36\pi$

d. $V(t) = 12\pi t^3 + 36\pi$

We are given that $\frac{dV}{dt} = 12\pi$. To find the volume $V(t)$, we need to integrate $\frac{dV}{dt}$ with respect to t. $V(t) = \int \frac{dV}{dt} dt = \int 12\pi dt$ $V(t) = 12\pi t + C$ We are given that the initial volume is 36π cm^3, which means $V(0) = 36\pi$. Substituting $t = 0$ into the equation for $V(t)$: $36\pi = 12\pi(0) + C$ $C = 36\pi$ Therefore, the volume of the balloon after t seconds is $V(t) = 12\pi t + 36\pi$.

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